LA HW

Included exercises: 1, 8, 14, 19, 21, 22, 27, 28, 29, 31, 32

Exercises 1–18: Invertibility and Inverses

Determine whether each matrix is invertible. If so, find its inverse.

Original Exercise 1

$$\begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}.$$

Original Exercise 8

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 5 \\ 1 & 3 & 1 \end{bmatrix}.$$

Original Exercise 14

$$\begin{bmatrix} 1 & -2 & 1 \\ 1 & 2 & -1 \\ 1 & 4 & -2 \end{bmatrix}.$$

Exercises 19–26: Computing $A^{-1}B$

Use the row-reduction algorithm to compute $A^{-1}B$.

Original Exercise 19

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 1 \end{bmatrix}.$$

Original Exercise 21

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 4 & 26 \\ 0 & -28 & -4 \end{bmatrix}.$$

Original Exercise 22

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 4 \\ 2 & -2 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & -2 \\ 1 & -1 \\ 4 & 2 \end{bmatrix}.$$

Exercises 27–34: RREF as Left Multiplication

For each matrix A, determine (a) its reduced row echelon form R and (b) an invertible matrix P such that $PA = R$.

Original Exercise 27

$$A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 1 & -1 \end{bmatrix}.$$

Original Exercise 28

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 2 \\ 1 & 0 & 1 \end{bmatrix}.$$

Original Exercise 29

$$A = \begin{bmatrix} -1 & 0 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 2 & 3 & -1 & -5 \end{bmatrix}.$$

Original Exercise 31

$$A = \begin{bmatrix} 2 & 1 & 0 & -2 \\ 0 & 1 & -1 & 0 \\ -1 & -2 & 2 & 1 \\ 1 & 3 & 1 & 0 \end{bmatrix}.$$

Original Exercise 32

$$A = \begin{bmatrix} 1 & -1 & 0 & -1 & 2 \\ -1 & 1 & 1 & -2 & 1 \\ 5 & -5 & -3 & 4 & 1 \end{bmatrix}.$$